

Nguyen Minh Thuan

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EDUCATION

SaiGon University
Information Technology

June 2024
Current GPA: 3.6/4.0

COURSEWORK

Courses: Data Structures and Algorithms, Object-Oriented Programming, Open Source Software, Calculus, Analysis and Design of Algorithms, Scientific Research Methods, Computer Architecture, Computer Networks

SKILLS

Languages: C/C++, C, PHP, Python, Java, JavaScript, HTML/CSS, LaTeX
Tools: Git/GitHub, NetBeans, VS Code, IntelliJ PyCharm/IDEA
Frameworks: React, SpringBoot
Libraries: pandas, NumPy, Matplotlib

PROJECTS

- SmartDoc AI – Intelligent Document Q&A System** | *Python, LangChain, FAISS, Qwen2.5:7b, Embedding, RAG* 2025
- Built an end-to-end RAG pipeline enabling users to upload PDF documents and ask questions in natural language
 - Integrated FAISS vector database to store and retrieve dense text embeddings with fast semantic similarity search
 - Deployed Qwen2.5:7b locally as the generative backbone, supporting Vietnamese and 50+ languages at zero API cost
 - Implemented automatic language detection and context-grounded answer generation to minimize hallucination
- Colorful Image Colorization** | *Python, PyTorch, NumPy, Deep CNN, CIE Lab, ImageNet* 2025
- Implemented a deep convolutional neural network to automatically colorize grayscale images, inspired by Zhang et al. (ECCV 2016)
 - Applied K-means clustering to quantize CIE Lab color space into 313 bins, framing colorization as a classification problem
 - Experimented with three colorization strategies: regression, classification-centroid, and softmax-weighted color averaging
 - Evaluated results quantitatively via MSE & PSNR and qualitatively via Visual Turing Test and side-by-side comparisons
- Feeding Frenzy – Big Fish Eats Small Fish** | *Python, Pygame, NumPy, OpenCV (cv2)* 2025
- Developed a full arcade game featuring fish size-based predation, leveling system, boss encounters, bombs, and experience stars
 - Implemented a dash mechanic and smooth physics-based movement using NumPy for vector calculations
 - Integrated real-time hand gesture control via OpenCV, allowing players to steer the fish using webcam hand movements
 - Supported dual input modes: keyboard (WASD / arrow keys) and computer-vision-based gesture tracking
- School Bus Management – Real-Time Fleet Tracking System** | *Node.js, Express, PostgreSQL, WebSocket* 2025
- Developed a low-latency tracking system using WebSockets to provide parents with real-time bus location updates
 - Integrated Mapbox API for precise route visualization and geographic notification triggers
 - Improved student safety by automating arrival alerts and real-time monitoring for administrators

EXPERIENCE

Academic Projects | *Student Developer* 2023 – Present
Independently built multiple software and AI projects throughout university, including a RAG-based Q&A system, an image colorization CNN, a real-time bus tracking system, and a computer-vision-controlled arcade game